

Ethics at the Front-End: Responsible User-Facing Design for AI Systems

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Abstract

AI ethics discourse typically centers on the design of algorithms and back-end systems, but the design of what users experience at the ‘front-end’ of AI systems also engages with values-laden ethical decisions. Deceptive patterns, distorted data visualization, and exclusionary interfaces represent unethical practices within the purview of interface, interaction and user experience design. In this workshop we seek to better understand and articulate what we believe to be an under-valued area of AI Ethics: front-end design. Through a combination of cross-disciplinary and collaborative hands-on activities, we aim to map a landscape of ethical front-end design for AI including an initial round-up of critical issues for practice, policy implications, and pressing areas for future research. The workshop will host a keynote by human-centered AI trailblazer, Ben Shneiderman and follow-up will include a written synthesis of outcomes for sharing with the HCI community.

CCS Concepts

• **Human-centered computing**; • **Interaction Design**; • **Interaction Design Process and Methods**;

Keywords

ethical design, AI ethics, human-computer interaction, deceptive design

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1 Motivation

The introduction of AI into all aspects of our lives has brought a rush of ethical issues into the foreground. Simultaneously, AI enables new kinds of user-facing design, for example, the dynamic tailoring of presentation, icons and interactions to an individual’s profile, search history or use patterns. Front-end design is also intimately entangled with ethically contentious AI practices like microtargeting, and it falls largely on the quality of front-end design



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to determine how to ethically handle critical elements like consent, privacy controls, and transparency. Although it's an algorithm that targets a user, a designer must decide what gets presented to that user to further persuade them how to think or act. Behavior design can be amplified and scaled by AI and AI can be aided and abetted by principles of behavioral design. Still the role of front-end design is undervalued within AI ethics discourse, and this workshop aims to help restore its place in the conversation as well as better articulate its boundaries, best practices and research priorities.

Herein, by front-end design we refer to user-facing elements, including interface and interaction design, as well as the user research that informs such design. By AI we refer broadly to the many systems incorporating advanced techniques like machine learning or multimodal processing for recognition, prediction or automation tasks, for example those that leverage computer vision, language or generative models. However, for formality, we defer to the ISO definition of AI as “computer systems capable of performing tasks that typically require human intelligence, such as reasoning, learning, perception and language understanding.” [32]

2 Background and objectives

If we understand AI Ethics as concerning “the psychological, social, and political impact of AI” [21] it's clear that front-end design must be taken seriously within this conversation. Work within HCI, outside the AI ethics space, but on ethical issues arising at the level of front-end design, has much to offer. For example, research on ‘dark’ or deceptive patterns [18], distorted data visualization [31], exclusionary interfaces [24] and even participation-washing [30] all represent unethical practices with relevance to AI systems. On the flip side, work on ‘bright’ patterns [2], inclusive design [20], privacy-preserving design [11, 16] front-end approaches to explainability/transparency [7] and meaningful co-design [29] can all contribute to a landscape of more ethically designed AI systems.

More broadly, work across HCI has also advanced various related areas of design ethics, from ethics education for designers [1, 2, 10] and ‘more-than-human’ ethics in design [15], to research on how design practitioners approach ethics in the workplace [22, 23, 28] which includes work on ethical design methods [9] and the role of ethics in research [8].

While questions of ethics in interface and interaction design have been active areas of HCI research, they have been somewhat sidelined in the broader debates around AI ethics. A glance at seminal AI Ethics handbooks reveals this notable absence. [4, 14] The field of AI Ethics has largely focused on the design of algorithms themselves and the harms that result from algorithmic influence on human decision-making in contexts such as access to public services or the administration of justice. This lack of clear attention to ethical front-end design has also shaped the way regulation of AI systems has been constructed; most of it centers on the development of models and the contexts of their deployment. For example, the EU's AI Act – a landmark piece of regulation now in force – addresses front-end design mainly through transparency requirements. In this context, transparency simply means that providers must ensure users are aware they are interacting with an AI system, unless it's already obvious from the nature of the interaction.

In this workshop, we aim to build a stronger foundation for bridging existing work on the ethics of interaction and interface design with ongoing debates in AI ethics, for example, in relation to explainability, deception and AI-targeting policy. What more should we expect from providers of AI systems beyond simply ensuring that users know they are engaging with an AI rather than a human? Should elements of interface design be prohibited, or made compulsory, to ensure that interactions are neither coercive nor manipulative? These questions are becoming ever more urgent as large language models transform the ways we interact with AI and interactions are increasingly mediated through language itself.

In this workshop we seek to better understand and articulate the current state of knowledge, best practice and research priorities for ethical front-end design for AI with a view to better equipping research, practice and policy. Through a combination of collaborative exploration and the collective sharing of cross-disciplinary expertise we aim to map a research landscape, including an initial round-up of its constituent components, critical issues and most pressing areas for future research; gather guidance for practice; and formulate policy considerations.

2.1 Objectives

The workshop has four primary objectives and coinciding outcomes. The first is to more clearly articulate and circumscribe the boundaries of ethical front-end design for AI through multidisciplinary consensus. This will entail drafting a map of the research landscape to guide researchers interested in advancing knowledge in the space as well as to identify gaps where research might be needed to better inform practice and policy. The second objective is to share examples of both good and bad practice in ethical front-end design for AI. As such, activities will be structured to elicit strengths and positive examples as well as concerns and challenges. The third objective is to consider the implications of front-end design practice for AI policy such as the EU AI Act and generate considerations for policymakers. For example, this may include considering how HCI researchers might contribute to appropriate risk assessment practices, and where back-end technical reviews and front-end analysis might meet to address the transparency requirements the AI Act promises. Finally, the workshop will begin to reify a community that is currently dispersed across fields and locations. By coming together and better articulating this field of research, we can share expertise, find synergies and strengthen efforts through cooperation. This will include how we might better network with other communities working in connected areas of technology and ethics. Thus, the workshop objectives can be summarized as:

- **Map the landscape** – Including key challenges, methods, existing areas of strengths and critical gaps for future research.
- **Scaffold the sharing of good and bad practice** – Generate insight for practice and also identify a way for continuing to do so systematically going forward.
- **Generate policy considerations** – consider the implications of front-end design for AI policy and generate insights for policymakers

Table 1: Workshop schedule

Activity	Min	Description	Activity type
Introductions and talks	60	Participants introduce themselves and either present their paper, or if organizers, summarize insights from their related research.	Lightning talks
Keynote by Ben Shneiderman	30	Ben Shneiderman, HCI pioneer in human-centered approaches to front-end design and AI, will deliver the workshop keynote.	Keynote
BREAK		BREAK	
Unpacking front-end design for: Research, Policy, and Practice	40	Based on interest and expertise, participants join one of three groups: Research, Policy or Practice. In parallel, each group generates ideas for four categories: 1. Big Issues/Concerns 2. Strengths/Good practice; 2. Challenges/Gaps 3. Needs/Recommendations	Ideation, first individually, then as Research/Practice/Policy group with clustering.
Mapping the landscape	35	As a large group, participants process and cluster findings from the previous activity to create a map of the landscape including where research is needed to inform policy and practice.	Affinity mapping based on outputs of previous activity.
Steps forward to better research and practice	15	Reflect on key insights/findings from workshop and next steps for: a) sharing best practice in future b) creating a community/linking to similar communities c) influencing research and policy.	Group discussion

- **Reify a community** with explicit shared objectives based on researchers working in various areas of ethical front-end design of AI to identify synergies, share expertise and strengthen efforts collectively.

3 Workshop Format and length

The workshop will have a duration of **two sessions** (90 minutes each, plus break). Activities are outlined in Table 1. This is a tentative schedule, and final details may be adapted to meet participant needs and opportunities. Ben Shneiderman has already agreed to deliver a keynote for the workshop.

4 Organizers

As organizers, we bring together expertise from across the design ethics space in HCI covering research topics as diverse as critical methods, law, wellbeing, policy, pedagogy, feminism and decoloniality.

Naseem Ahmadpour is an Associate Professor at The University of Sydney, where she leads the Affective Interactions research group. Her research contributes better understanding of care as an ethical obligation and approach in technology design, practice and pedagogy.

Rafael A. Calvo is Professor and Chair of Design Engineering at Imperial College London where he teaches responsible engineering, design and innovation. He is internationally recognized for his work in HCI and intelligent systems applying human-centered approaches to AI in areas of health, education, and wellbeing. His

books include the *Oxford Handbook of Affective Computing* and *Positive Computing* (MIT Press).

Sai Shruthi Chivukula is Assistant Professor at Pratt Institute, New York. Her research investigates how designers and HCI practitioners navigate ethical decision-making in real-world, ecologically mediated contexts, using critical design research methods. She translates these insights into practical and open-access ethics-focused methods and pedagogical tools that support ethical awareness, reflection, and action in design education and practice.

Christian Dindler is Associate professor and distinguished senior innovator at Aarhus University. His research has contributed to the theory and practice of participatory design [5] using this perspective to explore ethics in design practice [12, 25] and how to empower children to engage critically and curiously with technology [13].

Colin M. Gray is an Associate Professor at Indiana University, where they serve as program director for a graduate program in HCI/d. Their research has contributed foundational knowledge on dark patterns [18] and practice-led ethics [17, 33], supporting connections across design, law, and HCI [3, 19].

Tomasz Hollanek is an Assistant Research Professor at the University of Cambridge specializing in AI ethics and critical design. He applies critical design methods to AI development and governance, focusing on the ethics of AI ‘companions’ and translating ethical and regulatory requirements into practical tools and processes for designers. Forthcoming book: *Why AI Needs Critical Design* (Cambridge University Press).

Shaimaa Lazem is an Associate Research Professor at the City of Scientific Research and Technological Applications in Egypt.

Her work focuses on HCI in non-Western contexts and cultures. Her research interests include critical methods, decolonization and participatory design in HCI for ensuring technology design serves everyone effectively. She currently works on human-centered approaches to the design of Natural Language Processing (NLP) applications in Africa and has been supported by a Google 2020 Award for Inclusion Research and a Google AI 2021 Award.

Gizem Öz is a Postdoctoral Researcher on the Design Ethics in Practice project at Aarhus University, Department of Digital Design and Information Studies. She has a background in Industrial Design and experience researching design practices in diverse settings, including factories, craft workshops, and rural or small-scale contexts.

Dorian Peters is Assistant Professor in Ethical Design at the Institute for Technology and Humanity at the University of Cambridge. Her contributions to advancing research in user-facing design in HCI include a framework [27], measures [6] and set of research-based design heuristics [26] for wellbeing-supportive design. Her books include *Interface Design for Learning* and *Positive Computing* (MIT Press).

Nadia Piet is an independent researcher and designer, and co-founder/creative director of AIxDESIGN, a global community and decentralized research lab for critical, creative, and communal approaches to AI. Her project, Feminist UX of AI, uses participatory methods to reimagine human-AI interaction. She is researcher-in-residence at Netherlands Film Academy, mentor at New Inc, and faculty at ELISAVA Barcelona's MA in Design for Responsible AI.

5 Plans to publish

Position papers will be published on **arXiv** and made available collectively via a dedicated **workshop website** at www.desirableAI.com/frontend-ethics.

6 Accessibility

We aim to prioritize accessibility in design and distribution of workshop communications and materials. We don't currently require any special access tools but we will contact the chairs if needs arise based on participant access needs. We will reach out to participants ahead of the event to understand how we can best support any access requirements and their full involvement.

7 Offline materials

For those who miss the session or wish to review content, we will record and make available the lightning talks and keynote via the workshop website (subject to speaker consent).

8 After the workshop

We will synthesize outcomes of the workshop activities into a **collaborative paper** including findings and recommendations for the community. It is anticipated that this will define, describe and bring attention to the area as well as provide an initial map for a research landscape with examples of existing work that can inform the space. As a submission venue we will target the upcoming **2026 special issue** on Design Ethics in *Interacting with Computers*.

9 250-word Call for Participation

In this workshop we seek to better understand what we believe to be an under-valued area of AI Ethics: front-end design. Much ethics discourse revolves around the design of back-end systems, but the design of what users experience at the 'front-end' entails many values-laden decisions too. For example, "dark" patterns, distorted data visualization, participation-washing and exclusionary interfaces all fall within the purview of front-end design. Through cross-disciplinary brainstorming and collaborative activities, we aim to map a landscape of ethical user-facing design for AI including best practices, policy considerations, and pressing areas for future research. The workshop will also host a keynote by Ben Shneiderman.

Prospective participants should submit a 2-4 page short paper (excluding references and using the CHI submission template) presenting a project, work-in-progress or commentary that contributes to a deeper understanding of what ethical front-end design for AI is, what issues it raises, or how we can do it better. This might be a demonstration of ethical design in action, or an argument for why some issue should be considered. We're especially interested in examples of ethical design issues that are only possible because of AI (e.g. design for microtargeting, recommender systems, integrations of GenAI, front-end approaches to explainability) or that are especially relevant to AI-enhanced systems (e.g. the design of consent, transparency or privacy controls). At least one author per accepted paper must register and attend. Accepted papers will be published on arXiv and the workshop website. We expect 20-30 participants. For more information see: www.desirableAI.com/frontend-ethics.

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